MD SHIHAB SARKER

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Portfolio | GitHub | LinkedIn

Education

BRAC University 2022–2026
B.Sc. in Computer Science and Engineering | CGPA: 3.62/4.00

Rangpur Government College 2020
Higher Secondary Certificate (HSC) | GPA: 5.00/5.00

Nilphamari Government High School 2018
Secondary School Certificate (SSC) | GPA: 5.00/5.00

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++, SQL, PHP, 8086 Assembly

Web and databases: React, Next.js, Node.js, Express.js, Tailwind CSS, MongoDB, PostgreSQL, MySQL, Prisma, REST APIs

AI and machine learning: scikit-learn, TensorFlow/Keras, pandas, NumPy, RAG, BGE-M3, BM25, ChromaDB, Ollama

Tools: Git, GitHub, Jupyter/Google Colab, Docker Compose, VS Code

Undergraduate Thesis

Civic.ai – A Citizen Centric AI-Powered Government Service Assistant
BRAC University | Five-person team GitHub
Python, BGE-M3, ChromaDB, BM25, Ollama

- Developed a source-grounded assistant for birth registration, passport, and BRTA services using 174 prepared source units and 3,709 chunks.
- Combined dense and lexical retrieval, reranking, and evidence selection with local LLM generation over Bangla-English source material.
- Compared against a dense-only baseline on 30 saved questions, assessing retrieval, latency, and model-judged answer quality with documented limitations.

Selected Projects

KeyNest – Property Listing and Rental Platform

GitHub

MongoDB, Express.js, React, Node.js, Tailwind CSS, Redux

- Built a MERN property platform with listing management, search/filtering, reviews, and a mortgage-calculator interface.
- Implemented REST APIs, JWT and Google OAuth sign-in, protected user/admin routes, and Redux-managed authentication state.

Kothiq – Bangla Voice Prototype

GitHub

Python, FastAPI, Gemini, faster-whisper, Bengali MMS-TTS

- Built a turn-based Bangla/Banglish browser voice interface linking speech recognition, LLM responses, audio playback, and session history.
- Added interchangeable speech-recognition and synthesis providers for local inference, Colab services, and hosted APIs.

Multi-Class Text Classification

GitHub

Python, scikit-learn, TensorFlow/Keras; Natural Language Processing

- Compared 22 text-representation/model combinations across ten categories, using BoW, TF-IDF, GloVe, and Skip-gram with traditional and neural classifiers.
- Achieved macro F1 of 0.7109 with Skip-gram/GRU on 56,000 validation examples; compared per-class metrics and training times.

MiniVSFS – Filesystem Image Builder

GitHub

C, binary file I/O; Operating Systems

- Implemented utilities to create filesystem images and add files, managing superblocks, inodes, allocation bitmaps, and directory entries.
- Used 4,096-byte blocks, binary serialization, structure-size checks, and CRC32-based metadata checksums.

Guess The Word – 8086 Assembly Game

GitHub

8086 Assembly, DOS/BIOS interrupts, MASM/TASM-compatible syntax

- Built a two-player word-guessing game with five-letter words, five attempts, and feedback for correct, misplaced, and incorrect letters.
- Implemented input/output with DOS interrupts and a BIOS-based timer, using assembly macros for repeated printing and timing operations.